

1. Introduction

Ten states take almost all of India's foreign direct investment. Since the Department for Promotion of Industry and Internal Trade began reporting state-wise FDI equity figures in October 2019, those ten have absorbed close to 97 per cent of the national total (DPIIT 2024). A natural question is whether fiscal discipline is part of what separates them from the rest, and whether it sorts investment among them.

Two literatures speak to this and rarely cite each other. One asks how state fiscal policy affects growth: Trivedi and Rajmal (2011) and Panda and Sahay (2022) both report that fiscal deficits contract subsequent growth. A debt-sustainability literature (Misra, Gupta and Trivedi 2023, and NIPFP 2024) asks whether current borrowing paths are solvent, but stops before asking what those paths do to the investment a state subsequently receives. NITI Aayog (2025) ranks states on fiscal metrics in its Fiscal Health Index and validates that ranking against no outcome at all. Internationally, Blonigen (2005) surveys the empirical determinants of FDI and Feld, Köhler, Palhuca and Schaltegger (2024) find that fiscal decentralisation matters for where it locates, but the sub-national Indian case is untested.

This note reports what a ten-state, five-year panel can and cannot establish about that question, and the answer is more useful for its negative half. States carrying persistently lower public debt do receive more FDI, the association is large, and five estimators drawn from three families that share no distributional assumptions all recover it with the same sign. But the low-debt states in this sample are the western and southern ones and the high-debt states the northern and eastern ones; seventy-two per cent of the cross-state variance in debt lies between those regional groups. Once regional indicators enter, the debt coefficient is indistinguishable from zero. We report that confound as the finding rather than as a caveat, because it bounds what any panel of this shape can identify, and because the design that would settle the question follows directly from it.

2. Data

The panel is balanced: ten states over five fiscal years, 2019–20 to 2023–24, giving fifty state-year cells. The ten are Maharashtra, Karnataka, Gujarat, Tamil Nadu, Haryana, Telangana, Jharkhand, Rajasthan, West Bengal and Uttar Pradesh — the largest FDI recipients among the states the Sixteenth Finance Commission reports on. Delhi ranks fourth on cumulative inflow but is a union territory, absent from the Commission's annexures, so West Bengal enters in its place. Forty-eight of the fifty cells carry a usable FDI observation; the two that do not belong to Uttar Pradesh, whose DPIIT series begins only in FY 2021–22 and whose earlier cells are treated as missing rather than zero.

Four fiscal indicators, each a share of gross state domestic product, measure discipline: gross fiscal deficit, outstanding liabilities (debt), own tax revenue and capital expenditure. All are computed from the Sixteenth Finance Commission's Volume II annexures (Government of India 2025), which derive from CAG-audited State Finance Accounts. Capital expenditure is not reported directly and is derived from the identity that the gross fiscal deficit equals the revenue deficit plus capital outlay plus net lending, so it includes net lending; it is therefore a noisy proxy for productive spending, and Section 4 declines to interpret its coefficient. FDI is DPIIT's (2024) state-wise equity series

and GSDP is MoSPI's (2024) comparable 2011–12 series, from which nominal growth is computed year on year. Literacy and urbanisation from the 2011 Census (Office of the Registrar General and Census Commissioner 2011) enter the pooled specification only.

Table I reports the range check run before any estimation. One feature anticipates everything that follows: debt-to-GSDP spans 17.6 to 42.5 per cent, and that spread is almost entirely cross-sectional. Decomposing each regressor's sum of squares, 91.5 per cent of the variation in debt lies between states and 8.5 per cent within them, with a mean pairwise rank correlation of 0.91 between any two years. GSDP growth runs the other way entirely: 98.3 per cent of its variation is within states. Two consequences recur below. The within-state debt coefficient is identified off very little variation and should be expected to be fragile, and the pooled growth specification has almost no cross-state variation to explain.

Table I: Descriptive statistics.

Variable	<i>N</i>	Mean	Std. dev.	Min	Max
Fiscal deficit (% GSDP)	50	2.82	1.26	−0.7	5.4
Debt (% GSDP)	50	28.82	6.55	17.6	42.5
Own tax revenue (% GSDP)	50	6.47	0.80	5.2	8.2
Capital expenditure (% GSDP)	50	2.27	1.01	0.1	4.6
FDI (Rs crore)	48	31,194	43,933	44	163,795
GSDP growth (%)	50	10.39	7.35	−4.38	26.76

The two missing FDI cells are Uttar Pradesh in FY 2019–20 and FY 2020–21, for which DPIIT does not report separately.

3. Method

Two models are estimated. Model 1 takes log FDI as the dependent variable, Model 2 nominal GSDP growth, each on the four fiscal ratios and a COVID dummy:

$$\ln(\text{FDI}_{it}) = \alpha_i + \beta_1 \text{fd}_{it} + \beta_2 \text{debt}_{it} + \beta_3 \text{otr}_{it} + \beta_4 \text{capex}_{it} + \beta_5 \text{covid}_t + \varepsilon_{it} \quad (1)$$

Both are run twice: entity fixed effects with Driscoll and Kraay (1998) standard errors, and pooled OLS with the two Census controls. Keeping both separates the within-state from the between-state dimension, and that distinction turns out to carry the argument.

Ten clusters is far too few for the cluster-robust *t*-distribution to be reliable, so every coefficient the paper relies on is re-estimated by wild cluster bootstrap imposing the null, with Rademacher weights over the ten states (Cameron, Gelbach and Miller 2008). That bootstrap, not the cluster-robust *p*-value, is the arbiter throughout. We also report a Mundlak (1978) correlated random effects specification, in which each regressor enters twice — as the state's five-year mean and as the annual deviation from it — so that the between-state and within-state coefficients are identified jointly and can be tested against each other inside one equation.

Because the panel is short and one of its five years is an extreme common shock, the same prediction problem is run again through LASSO (Tibshirani 1996), Elastic Net (Zou and Hastie 2005) and SHAP values (Lundberg and Lee 2017) from a shallow random forest (Breiman 2001), under

state-blocked cross-validation. These stay credible at forty-eight observations where instrument-heavy panel GMM estimators do not (Roodman 2009, and Mullainathan and Spiess 2017). If a penalised linear selector, a tree-based explainer and a panel regression all land on the same variable, the agreement is worth more than any one p -value — subject to a qualification Section 5 makes precise.

One methodological failure is worth reporting because it is easy to repeat. An early machine-learning specification included the Census controls and ranked urbanisation the dominant predictor. Urbanisation is fixed within a state across all five years, so the forest was using it to identify which state an observation came from, re-learning the fixed effect rather than measuring agglomeration. Both controls are excluded from the machine-learning stage for that reason.

4. Results

4.1 Debt and FDI

The fixed-effects specification ($n = 48$, within $R^2 = 0.429$) and the pooled specification ($n = 48$, $R^2 = 0.643$) disagree about the sign of debt, and both are estimated precisely enough to take seriously (Table II). Pooled, debt is negative: a one-percentage-point lower debt-to-GSDP ratio is associated with roughly 19 per cent higher FDI, since $\exp(0.176) - 1 = 0.192$. Within states, it is positive.

Table II: Model 1, log FDI.

Variable	FE + Driscoll-Kraay		Pooled OLS + controls	
	Coef.	p	Coef.	p
Fiscal deficit	+0.566	0.000	+0.567	0.265
Debt-to-GSDP	+0.353	0.000	−0.176	0.007
Own tax revenue	+0.107	0.684	+0.325	0.268
Capital expenditure	−0.173	0.571	−0.366	0.008
COVID year	−1.406	0.000	+0.547	0.161
Literacy	—	—	+0.101	0.035
Urbanisation	—	—	+0.005	0.909

Literacy and urbanisation are time-invariant and absorbed by the fixed effects, so they enter the pooled specification only.

Table III re-estimates these under the wild cluster bootstrap, and the picture thins considerably. The pooled debt coefficient falls to $p = 0.057$, short of five per cent. Literacy and capital expenditure do not survive at all, and no claim rests on either. The within-state fiscal deficit coefficient, highly significant under Driscoll-Kraay, collapses to $p = 0.836$; no deficit-FDI claim is made anywhere in this note.

Table III: Cluster-robust and wild cluster bootstrap p -values.

Coefficient	Estimate	Cluster p	Bootstrap p	Sig. at 5%
M1 pooled: debt	−0.176	0.007	0.057	No
M1 pooled: literacy	+0.101	0.035	0.235	No
M1 pooled: capital expenditure	−0.366	0.008	0.095	No
M1 within: debt	+0.353	0.000	0.031	Yes
M1 within: fiscal deficit	+0.566	0.264	0.836	No
M2 within: debt	+1.097	0.001	0.020	Yes
M2 within: own tax revenue	+4.083	0.045	0.028	Yes
M2 within: COVID dummy	−13.783	0.000	0.003	Yes

The bootstrap imposes the null and uses Rademacher weights over the ten states, and is the appropriate reference distribution at this number of clusters.

4.2 Between states, not within them

The sign reversal is not an artefact of comparing two estimators with different assumptions. The Mundlak specification recovers it inside a single equation (Table IV), and the hypothesis that the within-state and between-state debt coefficients are equal is rejected at $p = 0.017$ under the same bootstrap. A between estimator fitted to the ten state means puts the cross-state coefficient at −0.237, so the pooled estimate is if anything conservative.

Table IV: Mundlak decomposition of the debt coefficient.

Specification	Coef.	p	What it identifies
Mundlak: within-state deviation	+0.370	0.0002	Movement around a state's own mean
Mundlak: state-mean term	−0.256	0.0001	Differences between state averages
Test of equality	0.626	0.017	The two dimensions differ
Between estimator	−0.237	0.022	Cross-state relation, direct

Standard errors clustered by state. The equality test is reported at its wild cluster bootstrap p -value.

The two coefficients answer different questions. The pooled one asks whether a state that persistently carries less debt than its peers attracts more investment than they do; it does. The within-state one asks whether a state's own FDI rises when its own debt rises above its usual level. Over this window it does, and not because borrowing attracts investors: deficits and debt rose everywhere at once in FY 2020–21 while FDI fell everywhere, and inflows then recovered while debt stayed at its elevated level. The within-state estimate describes the COVID window rather than a behavioural relationship, and Section 5 shows it does not survive a full set of year effects.

4.3 Machine-learning agreement, and growth

Both machine-learning families rank debt first among the five fiscal predictors by a wide margin (Table V), agreeing in sign with the pooled regression. Elastic Net is a close relative of LASSO rather than an independent test, and we count them as one verdict.

Table V: Machine-learning importance, five fiscal predictors.

Variable	LASSO	Elastic Net	Mean SHAP
Debt-to-GSDP	−1.376	−1.280	1.029
Fiscal deficit	+0.563	+0.488	0.219
Capital expenditure	−0.553	−0.562	0.136
Own tax revenue	+0.039	+0.098	0.208
COVID year	+0.129	+0.161	0.007

LASSO and Elastic Net coefficients are standardised, under state-blocked cross-validation. SHAP importance is the mean absolute SHAP value from a shallow random forest across all forty-eight observations.

The growth hypothesis fails outright. Under a single COVID dummy, within-state debt (+1.097) and own tax revenue (+4.083) both clear five per cent under the bootstrap. Under a full set of year fixed effects the own tax revenue coefficient vanishes (+0.896, $p = 0.611$) and the debt coefficient reverses sign (−0.689, $p = 0.086$). Nothing in the growth equation is stable across the two time controls, in either timing, and the pooled form has under two per cent of its dependent variable’s variation between states to explain in the first place. We report the growth hypothesis as unsupported by this panel.

4.4 Robustness and diagnostics

Two objections arrive before a referee has finished the results, and both can be answered from the panel. The first is that a five-year window containing a pandemic may be driving everything. The second is that ten states leave any estimate hostage to one of them. Table VI refits the within-state specification of Model 1 dropping, in turn, the COVID year, Jharkhand, and Uttar Pradesh.

Table VI: Model 1 within-state coefficients across subsamples.

Subsample	n	Fiscal deficit	Debt	Own tax rev.	Capital exp.
Baseline, all data	48	+0.566 (0.000)	+0.353 (0.000)	+0.107 (0.684)	−0.173 (0.571)
Excluding FY 2020–21	39	+0.802 (0.017)	+0.324 (0.002)	+0.535 (0.260)	−0.207 (0.480)
Excluding Jharkhand	43	−0.310 (0.141)	+0.182 (0.015)	+0.019 (0.922)	+0.414 (0.084)
Excluding Uttar Pradesh	45	+0.532 (0.000)	+0.366 (0.000)	+0.092 (0.711)	−0.186 (0.555)

Driscoll-Kraay p -values in parentheses.

Debt is the stable coefficient: +0.353 at baseline, +0.324 without the COVID year and +0.366 without Uttar Pradesh, and +0.182 without Jharkhand — smaller, but the same sign and still significant. That Uttar Pradesh changes almost nothing is the specific check its two censored cells invite. The fiscal deficit coefficient is not stable: dropping Jharkhand reverses its sign altogether. This is consistent with the bootstrap verdict of Table III, which already declines to credit a deficit effect, and it is why none is claimed here.

Table VII: Specification and diagnostic tests.

Test	Model 1 (log FDI)	Model 2 (growth)
F-test for poolability	6.741 ($p < 0.0001$)	0.818 ($p = 0.603$)
Breusch-Pagan LM, pooled vs. RE	0.397 ($p = 0.529$)	2.569 ($p = 0.109$)
Pesaran CD, cross-sectional dependence	0.019 ($p = 0.985$)	8.397 ($p < 0.0001$)
Wooldridge, first-order serial corr.	-0.051 ($p = 0.061$)	-0.092 ($p = 0.021$)
Modified Wald, groupwise heterosked.	347.33 ($p < 0.0001$)	184.58 ($p < 0.0001$)
Ramsey RESET, functional form	1.455 ($p = 0.246$)	—

The Hausman test is not computable for Model 1 and returns $p = 0.177$ for Model 2. Wooldridge tests $H_0 : \rho = -0.5$.

Three of these bear on how the estimates should be read. Groupwise heteroskedasticity is present in both models, which is why every standard error reported here is robust. Cross-sectional dependence and serial correlation are present in the growth equation and not in the FDI equation, which is among the reasons Section 4.3 declines to claim a growth result. And the poolability test rejects for Model 1 but not for Model 2, which is why the FDI specification carries state fixed effects while the growth specification gains little from them.

5. The regional confound

Group the ten states by region and the debt ordering is largely a regional ordering. Mean debt runs 20.6 per cent of GSDP in the west (Maharashtra, Gujarat), 26.2 in the south (Karnataka, Tamil Nadu, Telangana), 33.1 in the north (Haryana, Rajasthan, Uttar Pradesh) and 34.5 in the east (Jharkhand, West Bengal). Seventy-two per cent of the cross-state variance in debt lies between those four groups rather than within them. Regional indicators alone explain more of log FDI than debt alone does, at $R^2 = 0.60$ against 0.31.

Figure 1 shows the problem directly. The four groups separate along the debt axis almost without overlap: the two western states hold the low-debt, high-FDI corner, the three southern states sit beside them, and the northern and eastern states occupy the high-debt end. A line fitted through these ten points is a debt line and a region line at once, and nothing in the figure tells them apart.

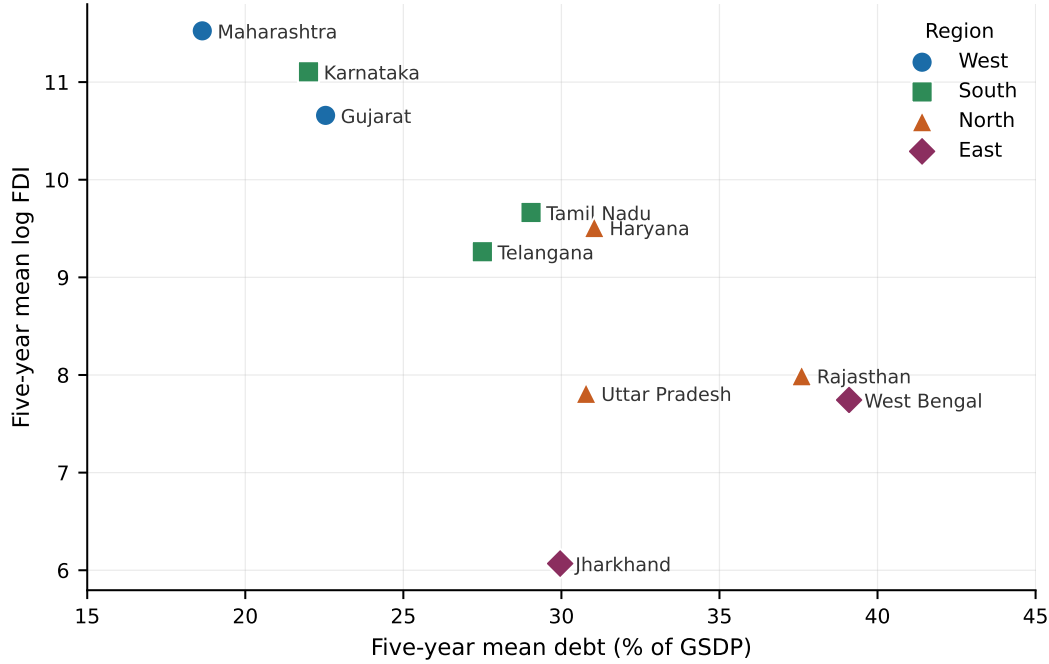


Figure 1: Five-year mean debt against five-year mean log FDI, by region.

Table VIII reports what happens to each of the paper’s estimates when regional indicators are added. The pooled coefficient reverses sign and becomes indistinguishable from zero. The Mundlak between-state term reverses. The rank correlation collapses. LASSO promotes a western-region indicator above debt. Within regions the ordering is not reliable either: West Bengal carries more debt than Jharkhand and receives more investment; Rajasthan carries more than Haryana and receives less.

Table VIII: The debt result with and without regional indicators.

Method	Statistic	Without region	With region
Pooled OLS	Debt coefficient	−0.176	+0.036
Mundlak	Between-state debt	−0.256	+0.151
Rank correlation	Spearman’s ρ	−0.842	−0.285
LASSO	Standardised debt coef.	−1.376	−0.679
Random forest	Mean SHAP , debt	1.029	1.006

Rank correlation is Spearman’s ρ between five-year mean debt and five-year total FDI across the ten states; “with region” removes regional means from both series. The four regions are west, south, north and east as grouped in the text.

Only the SHAP ranking is unmoved. That is the weakest of the five as a defence, because a regional indicator is constant within a state across all five years — exactly the property that made urbanisation unusable in Section 3 — so a tree can use it to identify states rather than to measure geography. The defence applies to the machine-learning results and not to the pooled regression or the rank correlation, and those are where the cross-state claim actually lives.

Two things follow. First, the convergence of five methods is not independent evidence against this confound: they converge partly by reading the same confounded cross-section. Second, the confound cannot be resolved by any refinement of this sample. With ten states and four regions, regional effects absorb three of the nine between-state degrees of freedom, and debt and region are close to collinear by construction. A reader who prefers the regional reading cannot be refuted from these data.

One further alternative deserves stating, because it pushes the same way. DPIIT records equity inflows against the state in which the recipient company is registered, not the state in which the investment is deployed. That error is not classical: it concentrates in the states hosting corporate headquarters, which here are Maharashtra, Karnataka and Gujarat, three of the four lowest-debt states in the panel. A registered-office effect would therefore bias the between-state coefficient in exactly the direction we report. The closest available test is a market-size control, since headquarters concentration follows market size above all: adding log GSDP to the pooled specification leaves the debt coefficient at -0.173 , essentially unmoved, while collapsing the literacy and capital expenditure coefficients, which are themselves scale proxies. That does not rule the channel out — a control for size is not a control for where a firm chooses to register — but it does mean the concern cannot be carried by state size alone.

That is a statement about the design rather than about the question, and the distinction matters because the Commission's annexures already contain the fiscal side of the remedy. The Sixteenth Finance Commission reports all four indicators for twenty-eight states and DPIIT reports a cumulative inflow for each. Across the ten states studied here, 72 per cent of the cross-sectional variance in debt lies between regions and the mean within-region spread is 6.7 percentage points. Across twenty-six states with both series available, the between-region share falls to 35 per cent and the mean within-region spread more than doubles to 15.1 points, with six regions averaging over four states each rather than four regions averaging two and a half. A cross-section on those twenty-six states points the same way as this panel: debt alone gives -0.299 ($p = 0.004$), and with regional indicators added it holds at -0.143 ($p = 0.056$) instead of collapsing to zero as it does here. We do not report that as a result — the wider set has no GSDP series in our hands and so no market-size control, and a cumulative total over a different window is not the annual outcome modelled here. We report it as evidence that the identifying design exists and is within reach.

6. Conclusion

Foreign direct investment in India is more likely where a state holds its debt down relative to its own economy, but only between states, not within a state from one year to the next. Five estimators from three families agree on the sign, and the between-within split survives inside a single equation. What none of them can do is separate that fiscal signal from a regional one. Ten states across four regions do not contain the variation required, and the honest report of this panel is the identification problem rather than the coefficient.

Two implications follow for practice. For a state finance department, the reading consistent with these estimates is that fiscal position operates as a structural, between-state signal rather than a lever: a state that has held debt below 25 per cent of GSDP for five consecutive years is in a different competitive tier from one that has not, and no single budget year moves that. For an index

builder — NITI Aayog (2025) being the immediate case — the result is a caution. The composite ranking constructed here is outperformed by debt alone on every outcome tested, and a composite validated against no outcome cannot know whether it is measuring anything.

Three limits bound all of this beyond the identification problem itself. The panel is short and one of its five years is an extreme common shock, so the within-state estimates describe that window as much as any behavioural relationship. Debt is the on-budget figure from the audited accounts and excludes off-budget guarantees, which several of these states use. And association is all a five-year panel of ten states can support: reverse causality, in which anticipated investment relaxes a state's borrowing constraint rather than the reverse, is not tested here and cannot be ruled out.

The extension that would settle the question is specific rather than aspirational: a panel wide enough to hold region fixed while debt still varies within it. Twenty-six states clear that bar where ten do not.

Data and code availability

The dataset, the analysis code, and the full-length working paper from which this note is drawn are openly available at

<https://github.com/fbivinay/Fiscal-Discipline-and-Investment-Outcomes>

The repository holds the audited source series, the fifty-row analysis panel behind every estimate reported here, and every analysis script.

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